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# Graphical Abstract

## Learning Across Modalities: A Systematic Survey of Multimodal Models for Financial Analysis

Simone D'Amico, Fabio Mercorio, Navid Nobani, Giancarlo Sperli, Carmine Ventre

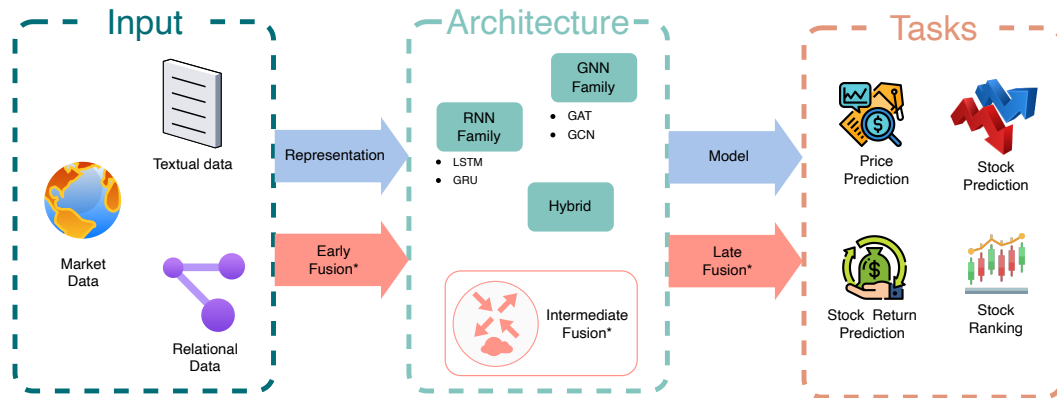


Figure 1: General flow of the framework: Different input modalities are processed to produce models capable of performing various tasks. Elements marked with \* indicate optional components.

# Highlights

## **Learning Across Modalities: A Systematic Survey of Multimodal Models for Financial Analysis**

Simone D’Amico, Fabio Mercorio, Navid Nobani, Giancarlo Sperli, Carmine Ventre

- Taxonomy for multimodal forecasting by modality, model, fusion type, and task.
- Review of 35 papers (2018–2025) analysed through the proposed unified taxonomy.
- Challenges: misalignment, modality imbalance, noisy inputs and generalization.
- Trends: adaptive fusion, missing modality learning, LLMs, temporal GNN models.

# Learning Across Modalities: A Systematic Survey of Multimodal Models for Financial Analysis

Simone D'Amico<sup>a,\*</sup>, Fabio Mercorio<sup>b</sup>, Navid Nobani<sup>b</sup>, Giancarlo Sperli<sup>a</sup>, Carmine Ventre<sup>c</sup>

<sup>a</sup>*Department of Electrical and Information Technology, University of Naples Federico II, Naples, Italy*  
*email: firstname.lastname@unina.it*

<sup>b</sup>*Department of Statistics and Quantitative Methods, University of Milan-Bicocca, Milan, Italy*  
*email: firstname.lastname@unimib.it*

<sup>c</sup>*King's College London, United Kingdom*  
*email: carmine.ventre@kcl.ac.uk*

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## Abstract

Multimodal learning has recently emerged as a powerful paradigm for financial forecasting, enabling the integration of heterogeneous data sources such as market time series, textual news, and relational graphs. This survey presents a unified taxonomy for multimodal financial forecasting models, structured along four key dimensions: input modalities, modelling architectures, fusion strategies, and predictive tasks. Using this taxonomy, we conduct a systematic review of 35 representative works published between 2018 and 2025, highlighting methodological trends, design choices, and performance patterns. Our analysis identifies persistent challenges, including temporal misalignment, modality imbalance, missing or noisy data, and limited cross-market generalization. We also discuss emerging trends and promising research directions, such as adaptive fusion, incomplete modality learning, and the integration of large language models and temporal graph neural networks, and analyse how architectural and fusion design choices impact practical considerations such as interpretability and deployability, aiming to bridge methodological innovation with domain-specific requirements.

**Keywords:** Fusion Strategies, Multimodal Analysis, Market predictive analysis, Stock Market analysis

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\*Corresponding author

## 1. Introduction

Financial forecasting has long been a central focus in academic research and industry, predicting market trends, asset returns, and risk dynamics to inform trading strategies and investment decisions. Traditional models primarily rely on historical market data, such as price and volume time series, to capture patterns and generate predictions [1, 2]. However, financial markets are also shaped by exogenous factors, including macroeconomic signals, geopolitical events, investor sentiment, and corporate disclosures, which cannot be fully captured by quantitative indicators alone [3, 4].

The growing availability of heterogeneous financial data, spanning structured inputs (e.g., financial statements and technical indicators) and unstructured sources (e.g., news, regulatory filings, and social media), has driven the adoption of multimodal approaches that integrate diverse data streams to better model market complexity. This shift has spurred interest in multimodal approaches, which aim to integrate heterogeneous data streams to better reflect the complex nature of financial systems. Recent studies demonstrate the effectiveness of multimodal fusion for stock prediction across sectors [5, 6, 7], with advanced methods incorporating causal reasoning and saliency-aware augmentation to improve robustness and exploit macroeconomic signals [8, 9].

Recent literature emphasizes the importance of integrating multiple sources of information to improve predictive performance and model robustness [10]. Additionally, hierarchical fusion strategies and large-scale neural network-based frameworks have been shown to enhance both predictive accuracy and interpretability [11, 12]. These sources include not only structured time series but also textual news, graph-based relationships, and alternative data streams.

The rise of deep learning has further accelerated this trend by enabling architectures that extract modality-specific features and combine them in flexible and scalable ways. As a result, financial forecasting has become an increasingly multimodal task, integrating temporal models (e.g., LSTMs [13]), language models (e.g., BERT [14]), relational architectures (e.g., GNNs [15]), and fusion mechanisms that align these representations within a unified decision framework. Recent works highlight the systematic review of multimodal fusion techniques and knowledge graph reasoning as emerging foundations for financial forecasting [16, 17, 18, 19]

### 1.1. Motivation and Contribution

Multimodal learning has gained prominence in financial forecasting by integrating heterogeneous sources such as time-series data, textual news, social media sentiment, and fundamental indicators. While each modality provides independent predictive signals, their combination yields richer representations of market dynamics, and empirical evidence shows that multimodal models often outperform unimodal approaches in tasks including stock movement prediction, risk assessment, and volatility estimation [20, 21, 22].

Despite this progress, the literature remains fragmented across architectural choices, data modalities, fusion strategies, and evaluation settings [23, 24]. Existing surveys provide valuable but focused perspectives: Wang et al. [23] review graph neural network methods and their financial applications, while Patel et al. [25] focus on graph-based prediction models for stock market forecasting, highlighting key results and open issues.

However, these works primarily address a single architectural paradigm and lack a systemic multimodal perspective. A unified taxonomy spanning data modalities (market, textual, relational), model architectures (RNNs, GNNs, Transformers, hybrids), fusion strategies (early, intermediate, late), and predictive tasks is still missing, limiting cross-model comparison and the identification of shared design principles.

To address this gap, this survey introduces a novel taxonomic framework that organizes state-of-the-art multimodal forecasting models along these dimensions. Beyond cataloguing existing work, we analyze methodological foundations, highlight recurring challenges, and identify emerging research directions, providing a structured reference for both researchers and practitioners.

**Contributions.** We aim to clarify how heterogeneous data sources are integrated across existing works, which architectural solutions are most commonly adopted, and which challenges remain open in this rapidly evolving field. Based on this motivation, we summarize our main contributions as follows:

1. **Taxonomy-Based Analysis:** We propose a novel taxonomy that organizes multimodal forecasting models along four dimensions: input modalities, model architectures, fusion strategies, and predictive tasks. The taxonomy, presented in

Section 3, enables consistent comparison and highlights recurring design patterns and trade-offs.

2. **Systematic Literature Review:** A comprehensive review of 35 papers published between 2018 and 2025, analyzed through the proposed taxonomy and covering diverse data sources, architectures, and forecasting objectives.
3. **Insights and Research Directions:** A synthesis of key technical challenges—such as temporal misalignment, missing or noisy data, and modality imbalance discussed in Section 4 and outline potential directions for future research.

**Scope** We conducted a structured literature search on Google Scholar<sup>1</sup> targeting recent advances in multimodal models for stock market forecasting. The search combined domain-specific keywords to capture a broad set of works on multimodal representation and relational modeling.

The queries included terms such as *"multimodal model for stock forecasting"*, *"multimodality graph model for financial forecasting"*, and *"fusion input for financial forecasting model"*. In this way, we enabled the retrieval of relevant multimodal frameworks despite variations in terminology.

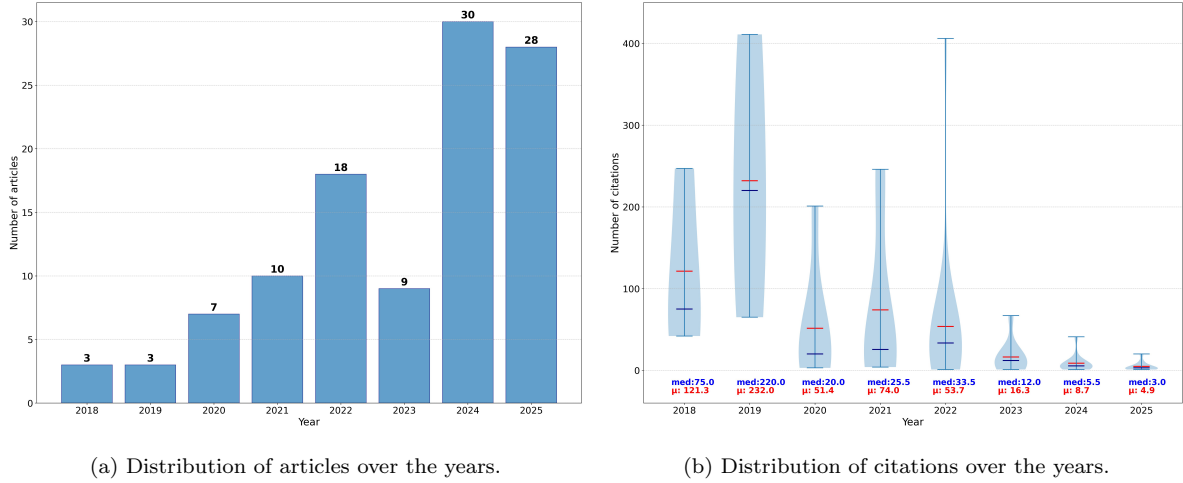


Figure 2: Analysis of the distribution of papers by year and citation count.

Overall, the search identified 108 papers published between 2018 and 2025, including journal articles, conference papers, and influential arXiv preprints. Fig. 2a shows the

<sup>1</sup><https://scholar.google.com/>

temporal distribution of publications, highlighting strong recent growth with a slight dip in 2023, while Fig. 2b reports the citation distribution over time.

Selected journal papers are primarily from Q1-ranked venues, with a limited number of Q2 exceptions, while conference papers originate from A-ranked venues. A small number of non-peer-reviewed preprints were included due to their relevance.

The selected time frame reflects the emergence of enabling technologies, such as graph neural networks (2016–2017) and transformer-based language models (e.g., BERT, 2017), which catalyzed multimodal approaches in financial forecasting. Their integration with structured market data and alternative modalities (e.g., news and social signals) laid the foundation for recent multimodal models, making this period representative of both their emergence and evolution.

Papers were included based on two criteria: (i) the use of multimodal models combining at least two input data types beyond the target variable (e.g., market data with textual or event-based information); and (ii) application to stock prediction tasks such as price forecasting, movement classification, or return estimation. After screening, 35 papers met these criteria and form the basis of our review.

## 2. Definitions and Background

Multimodal models are systems designed to process and integrate data from multiple heterogeneous sources or modalities, such as text, time series, images, or graphs. In the context of financial forecasting, they aim to leverage diverse information—such as historical stock prices, financial news, social media sentiment, and inter-company relationships—to enhance predictive accuracy and model robustness.

Multimodal models have been widely studied in other domains, such as fake news detection [26], where surveys examine cross-modal interaction and fusion strategies. In contrast, this survey focuses on financial forecasting, where multimodal fusion raises domain-specific challenges related to temporal dynamics, market heterogeneity, and data availability.

Underlying these models is the core hypothesis that different modalities provide complementary perspectives [27, 28]. For instance, while numerical market data captures quantitative price movements, textual sources (e.g., news articles or social media)

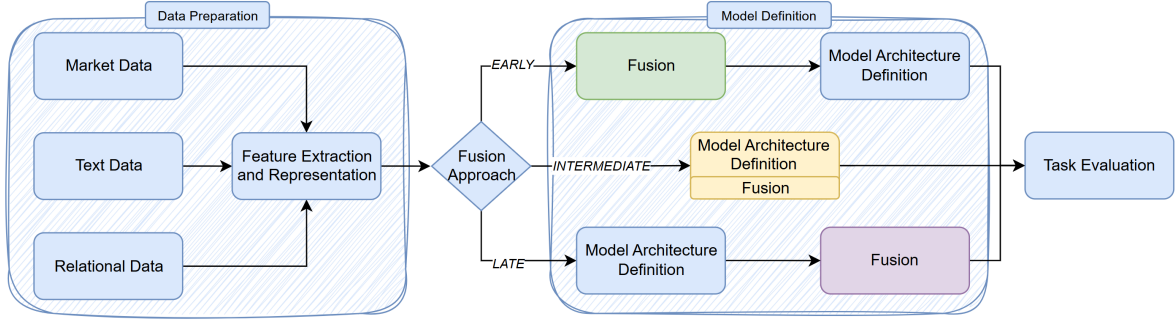


Figure 3: Workflow of a multimodal financial forecasting pipeline, showing data preprocessing, exclusive selection of fusion approach (early, hybrid, late), model definition, and evaluation.

offer insights into sentiment and external events that may not yet be reflected in market prices. In parallel, graph-based structures can model static or dynamic relationships between entities—such as industry dependencies or supply chain interactions—adding a valuable relational dimension. By combining these perspectives, multimodal models are able to generate more informed and context-aware predictions.

Figure 3 illustrates the general flow underlying multimodal models for financial prediction, structured into four key components: **Input**, **Architecture**, **Fusion**, and **Tasks**. These components represent the fundamental stages involved in designing and evaluating multimodal systems.

**Input:** Multimodal models typically process heterogeneous data sources, including market data (e.g., stock prices), textual data (e.g., financial news or social media), and relational or structured data (e.g., graphs of company relationships or supply chain relationships). This diversity of inputs reflects the real-world complexity of financial markets and is essential for capturing complementary signals.

**Architecture:** Once the data are represented in a usable format, different neural architectures are applied. These may include feedforward neural networks (FNN), recurrent neural networks (RNN) or graph neural networks (GNNs). Hybrid approaches that combine multiple architecture families are also increasingly adopted.

**Fusion:** A critical design choice in multimodal modelling is how and when to integrate different modalities. As Figure 3 shows, fusion strategies can be applied at various stages—early, intermediate, or late—each with different implications for model capacity and interpretability.

**Tasks:** The final goal of these models is to perform downstream financial prediction

tasks, such as price prediction, stock movement classification, pattern recognition, or stock ranking. These tasks guide the evaluation and benchmarking of the multimodal models.

This conceptual framework guides the structure of our survey: in this section, we introduce each of these four components in detail, providing the necessary technical foundations. Subsequently, in Section 3, we use this framework to systematically compare and categorise analysed papers.

### *2.1. Input Modalities*

Multimodal financial forecasting models exploit heterogeneous data sources to capture the multifaceted nature of financial markets. When properly integrated, these modalities provide complementary information that can enhance prediction accuracy and robustness. This section outlines the main data categories used in this context.

#### *Market Data*

Market data comprises time-series signals reflecting asset trading activity. The rationale is that prices often exhibit recurring patterns and cyclical behaviours that machine learning models can exploit. Market data can be broadly divided into raw trading data and technical indicators. Raw trading data typically includes Open, High, Low, and Close (OHLC) prices and traded volumes, which are available for virtually all publicly traded assets. Figure 4 illustrates a typical representation of these values.

Technical indicators, on the other hand, are mathematical transformations or summaries derived from OHLC data. Common examples include Moving Averages (SMA, EMA), Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Bollinger Bands, and stochastic oscillators [25]. These indicators are designed to capture patterns related to trend strength, momentum, volatility, and mean reversion.

In high-frequency trading or intraday forecasting settings, more granular information is often incorporated, such as bid-ask spreads, order book dynamics, or tick-level price updates. While time granularity can range from milliseconds to daily intervals, daily frequency remains the most common in academic literature due to data availability and computational constraints.

From a modelling perspective, traditional statistical methods like ARIMA [1] and



Figure 4: Candlestick chart for Apple Inc. (AAPL), displaying daily OHLC prices over a 12 months period. Green and red candles respectively indicate upward and downward movements in closing prices. The chart includes 5-day and 20-day moving averages to highlight short- and medium-term trends. At the bottom, volume bars show daily trading volume, offering insight into investor activity and market pressure.

GARCH [29] have been widely used on univariate price series. However, deep learning approaches—such as LSTM networks or Transformer-based architectures—are better suited to capturing complex, multivariate, and non-linear temporal dependencies [30].

**How to represent market data** To integrate market data into multimodal architectures, representations vary by abstraction level and task. A simple approach concatenates raw OHLC values into a single input vector, for instance by flattening a fixed-length historical window, which preserves temporal structure and suits models such as MLPs, LSTMs, or Transformers [31].

More structured representations compute statistical or technical features over pre-defined intervals. Indicators such as moving averages, volatility bands, or momentum scores reduce dimensionality while providing domain-specific interpretability, making them common in both traditional finance and deep learning [25, 23].

Alternatively, time series can be transformed into graph structures to capture in-

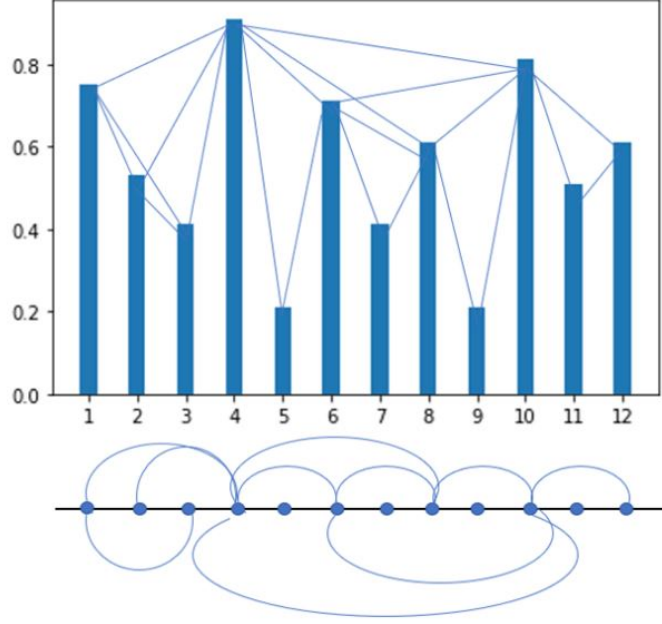


Figure 5: An example of the application of the visibility algorithm from [34].

intrinsic geometry and temporal relations. A prominent example is the Visibility Graph Algorithm, widely used in financial analysis (e.g.[32, 33]) because it is parameter-free and maps time series to scale-free graphs[34]. Each time point becomes a node, and edges connect points that satisfy a visibility criterion (Figure 5), enabling the extraction of topological features for graph-based or hybrid models.

Upon the definition given by Lacasa et al. [34], we can establish the following visibility criteria: two arbitrary data values  $(t_a, y_a)$  and  $(t_b, y_b)$  will have visibility, and consequently will become two connected nodes of the associated graph, if any other data  $(t_c, y_c)$  placed between them fulfils  $y_c < y_b + (y_a - y_b) \frac{t_b - t_c}{t_b - t_a}$ .

These representations—ranging from raw sequences to engineered features and graph-based encodings—can be seamlessly integrated with other modalities (e.g., textual news, images, or social signals) to capture the multifaceted nature of financial markets in multimodal forecasting tasks.

### *Textual Data*

Textual data can be grouped by source and intent. A first category comprises institutional and journalistic news, such as content from Bloomberg, Reuters, CNBC, the Financial Times, and Yahoo Finance, which report on earnings, mergers and ac-

quisitions, regulatory changes, and geopolitical events. Headlines and full texts can be analysed with NLP techniques to assess market relevance or extract sentiment signals.

A second category includes regulatory and official documents, such as central bank statements (e.g., FOMC<sup>2</sup> announcements) and corporate disclosures. Although highly structured and formal, these texts convey critical information for financial analysis.

A third category consists of social media platforms, including X<sup>3</sup>, Reddit<sup>4</sup>, and StockTwits<sup>5</sup>. These sources provide real-time insights into retail investor sentiment, rumours, and emerging narratives and, despite their noise, have demonstrated predictive value, particularly for retail-driven assets.

**How to represent textual data** In the context of multimodal models, representing textual data effectively is essential for integrating linguistic information with other modalities such as time series, graphs, or structured tabular inputs.

To extract informative signals from textual documents, approaches commonly rely on techniques such as tone analysis and keyword matching, often combined with contextual representations derived from BERT-based language models, which encode text into dense semantic vectors that can be combined with other modalities through attention mechanisms or vector concatenation. Features commonly extracted from social media include message volume, sentiment scores, and topic clusters.

Alternatively, linguistic feature engineering can be applied to extract interpretable information from text, such as domain-specific keywords or topic distributions using techniques like Latent Dirichlet Allocation (LDA) [35]. These features offer compact, high-level summaries of the underlying text and can serve either as direct input to the model or as a form of weak supervision [36].

Furthermore, sentiment analysis and emotion detection can provide additional signals by capturing subjective aspects of the text. These can be obtained using pre-trained models like FinBERT or lexicon-based approaches such as those proposed by Loughran and McDonald [37], yielding sentiment polarity scores or emotion labels that enhance

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<sup>2</sup><https://www.federalreserve.gov/monetarypolicy/fomc.htm>

<sup>3</sup><https://x.com/>

<sup>4</sup><https://www.reddit.com/>

<sup>5</sup><https://stocktwits.com/>

the model’s ability to interpret market mood.

Overall, textual information can be represented through dense embeddings, symbolic features, topic-based summaries, or affective scores, depending on the nature of the task and its interaction with other data modalities.

### *Relational data*

Relational data play a crucial role in stock prediction models, as financial markets operate as highly interconnected systems in which the behaviour of a single asset is often influenced by its relationships with other entities. Unlike market-based indicators or textual sentiment, relational data encode structural dependencies and latent correlations among firms, industries, and broader economic ecosystems.

One of the most common forms of relational information is a company’s **industry or sector classification**. Sector membership provides essential contextual information, as firms operating within the same sector frequently respond in similar ways to macroeconomic shocks, regulatory changes, or commodity price fluctuations. As a result, strong correlations in stock movements are often observed among sector peers. For example, a decline in oil prices may negatively affect multiple energy-sector companies simultaneously, regardless of firm-specific fundamentals.

Another important category of relational data concerns **corporate governance and ownership structures**. Information about board members, executives, and institutional investors can reveal non-trivial connections between companies, particularly when individuals or entities hold positions or stakes across multiple firms. Such links may influence investor expectations and market dynamics. For instance, the appointment of a well-regarded executive can positively affect market sentiment based on prior performance, while shared directors or cross-ownership arrangements may induce indirect dependencies that are not apparent from price data alone.

Finally, **supply chain relationships and strategic partnerships** represent a critical source of relational information. Disruptions affecting a key supplier can propagate through the supply network, impacting the financial performance of downstream firms. Explicitly modelling these dependency networks allows prediction models to capture the transmission of shocks across interconnected entities. Geographic relationships, such as shared operational regions or regional exposure, further enrich relational repre-

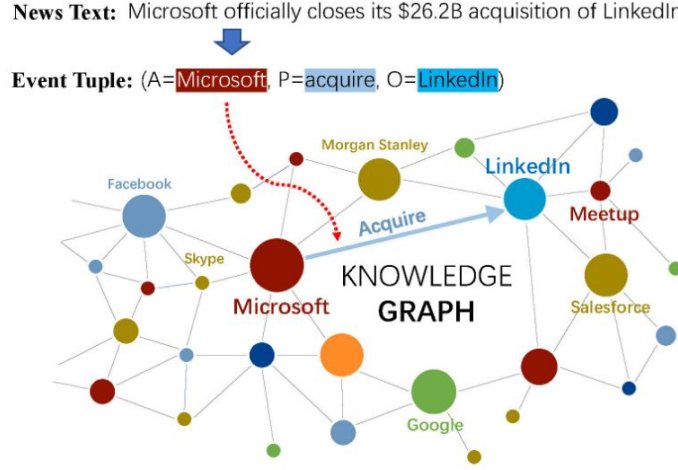


Figure 6: An example taken from [38] illustrates the analysis of associated events and how they affect relationships between stocks.

sentations, particularly in the presence of localised disruptions or geopolitical events.

**How to Represent Relational Data** By their nature, these complex relationships are best modelled as graphs, where nodes represent entities such as companies, sectors, or locations, and edges capture various types of relations, including ownership, collaboration, or similarity. From these comprehensive graphs, specific components of interest are often extracted to focus analysis. For example, the correlation network between stocks highlights pairs or groups of assets with strong price co-movements, which can inform portfolio diversification or risk assessment. Other sub-graphs might represent supply chain clusters or governance communities, enabling targeted analysis of propagation effects or influence patterns.

Events extracted from textual sources constitute another valuable form of relational data, as illustrated in Figure 6. Leveraging various event detection methods proposed in the literature [39], such information can be incorporated either as static features or as temporal profiles (e.g., updated quarterly). These features are frequently used as node attributes or edge weights within financial knowledge graphs and graph neural networks (GNNs), enabling more sophisticated relational reasoning [10, 23]

Overall, graph-based modelling of relational data provides a powerful framework to incorporate structural dependencies into financial prediction models, complementing traditional time series and textual data with rich, interpretable relational context.

## 2.2. Model Architectures

In multimodal financial forecasting, different neural network architectures serve as fundamental building blocks, each specialized in handling specific data types. These models—ranging from recurrent and convolutional networks to attention-based and graph-based architectures—are often individually applied to one modality (e.g., time series, textual information, relational data), and subsequently integrated into broader multimodal frameworks. Their modular design facilitates the extraction of modality-specific representations, which are then fused to capture cross-modal interactions and improve forecasting performance. This section presents the most widely used neural architectures in the financial forecasting literature, discussing their mechanisms, strengths, and typical applications.

### *Recurrent Architectures: LSTM and GRU*

**RNNs** have been widely adopted for modelling sequential data, particularly in the financial domain where time-dependent structures are critical. However, traditional RNNs often struggle to capture long-range dependencies due to the vanishing gradient problem. To overcome this limitation, more advanced architectures such as Long Short-Term Memory (LSTM) networks [13] and Gated Recurrent Units (GRU) [40] were introduced. These models have become fundamental components in financial forecasting tasks, including the prediction of stock prices, returns, and volatility [41, 42].

LSTM networks incorporate a gating mechanism that allows them to retain information over extended sequences, making them well-suited to capture temporal dependencies in noisy and non-stationary financial time series. This capability is particularly important in stock markets, where delayed reactions, autocorrelations, and irregular patterns are common. By selectively preserving relevant past information through memory cells and forget gates, LSTMs have proven effective in a variety of forecasting tasks [43].

### *Graph Neural Networks for Financial Forecasting*

Graph Neural Networks (GNNs) [44] have emerged as a powerful paradigm for modelling structured relational data. In financial forecasting, they are particularly effective for capturing the complex interdependencies among stocks, firms, or sec-

tors—relationships that are often encoded as graphs based on price correlations, supply chain links, industry classifications, or co-movement patterns.

In multimodal contexts, GNNs are frequently used alongside other models to integrate structured market signals (e.g., historical prices) with unstructured information sources such as news articles or social media content [23]. Each node in the graph typically represents a financial entity (e.g., a stock), while node features may include numerical descriptors, textual embeddings, or multimodal vectors. By propagating information through graph edges, GNNs allow joint reasoning over individual stock attributes and their relational context [45, 46].

Formally, the financial market can be represented as a graph  $G = (V, E)$ , where  $V$  denotes the set of nodes (stocks) and  $E$  the set of edges (relations). Given initial node features  $X \in \mathbb{R}^{|V| \times d}$ , GNNs iteratively update the representation of each node  $v_i$  by aggregating information from its neighbours  $\mathcal{N}(v_i)$ , enabling the model to capture both local interactions and broader market structure.

Among the various GNN architectures, two have been particularly influential in financial applications: Graph Convolutional Networks (GCN) [15] and Graph Attention Networks (GAT) [47].

**Graph Convolutional Networks (GCNs)** extend the notion of convolution to graph-structured data by implementing localized message passing based on graph topology. In this framework, each node representation is iteratively updated by aggregating information from its neighbours, leveraging a normalized adjacency structure. This mechanism allows the network to learn expressive node features that capture both local and global structural patterns.

**Graph Attention Networks (GATs)** enhance GCNs by introducing an attention mechanism that enables the model to assign different levels of importance to neighbouring nodes. This is particularly relevant in financial markets, where not all inter-stock connections are equally informative for price movements. Instead of uniformly averaging neighbour features, GATs learn to focus on the most influential relationships.

Overall, GNNs provide a natural and flexible architecture for multimodal fusion. While Transformer-based models are effective at capturing modality-specific information (e.g., language or price patterns), GNNs excel at learning structured dependencies

among financial entities. Recent work has begun integrating GNNs, particularly GATs, with other deep learning backbones such as LSTM or BERT, enabling joint learning over textual signals, temporal dynamics, and graph-structured relations [45].

### 2.3. Fusion Approach

In multimodal learning, fusion mechanisms refer to the strategies used to combine information from different input modalities—such as text, time series or graphs—into a unified representation suitable for prediction tasks. The choice of fusion strategy plays a crucial role in the model’s ability to capture complementary information and exploit cross-modal interactions. Broadly, fusion mechanisms can be classified into three categories: early fusion, late fusion, and attention-based (or intermediate) fusion. Each of these strategies reflects a different design philosophy and offers specific trade-offs in terms of flexibility, complexity, and ability to model cross-modal interactions.

#### Early fusion

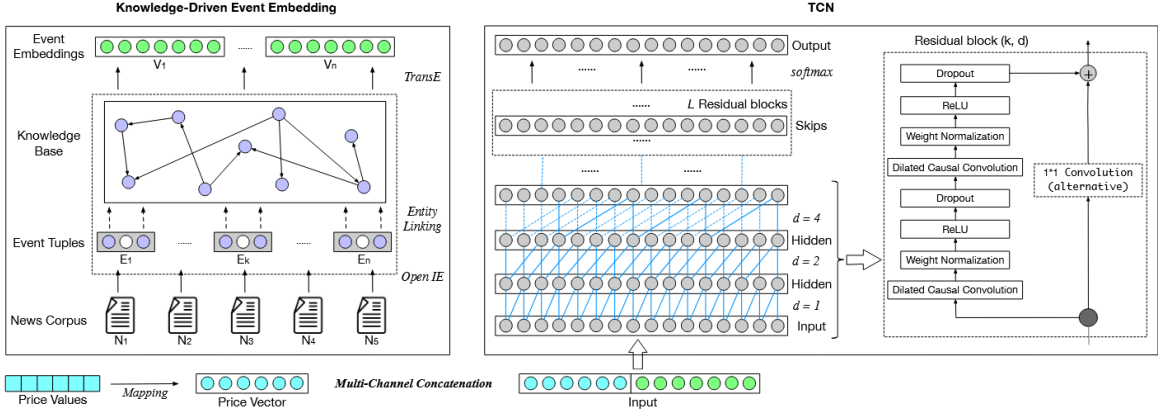


Figure 7: Example of early fusion taken from [48]. Following the vector representation of price values and events, the corresponding vectors are concatenated to produce a unified input to the KDTCN framework.

It refers to the combination of raw or low-level features from each modality into a single input representation, prior to any modelling. For example, one might concatenate a vector representing the sentiment of a news headline (extracted via a language model like BERT) with statistical indicators from financial time series (such as moving averages or volatility measures), and feed the resulting feature vector into a neural model

like an LSTM or Transformer. This method allows the model to learn joint representations from the start and can capture interactions between modalities at the feature level. However, early fusion assumes that all modalities are temporally or structurally aligned, and may suffer when there are large discrepancies in scale, reliability, or noise. Additionally, when input dimensions are large, early fusion can lead to over-fitting due to high model complexity.

### Intermediate fusion

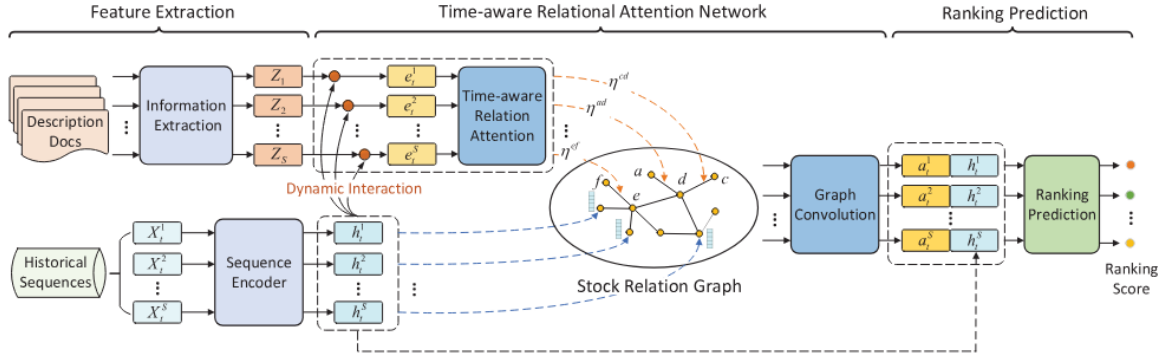


Figure 8: Example of intermediate fusion from [49]. The fusion of textual data and historical market data is performed through an attention mechanism to obtain representations enriched with relational features for all stock nodes.

Intermediate fusion provides a flexible and powerful mechanism to integrate different modalities during the representation learning phase. Rather than combining inputs at the feature level (as in early fusion) or predictions at the output level (as in late fusion), intermediate-base fusion methods allow a modality to dynamically attend to another based on relevance, enabling fine-grained, context-dependent interactions. One of the most relevant intermediate fusion mechanisms is based on the *attention mechanism* [50], which computes a weighted sum over a set of values based on the similarity between a query and each key.

From an information fusion perspective, attention-based fusion mechanisms naturally support conflict resolution and relevance filtering, by dynamically modulating the contribution of each modality when signals are noisy, sparse, or contradictory. At the same time, they facilitate the exploitation of modality complementarity by selectively emphasizing informative cross-modal interactions.

This allows the model to focus on the most relevant parts of one modality given the context of another. Attention-based fusion has been used in models that combine BERT-style textual encoders with LSTMs or GNNs that process temporal or relational data [45, 51]. Such models are particularly effective in financial settings, where modalities are often asynchronous, heterogeneous, and loosely aligned, and where capturing inter-modality dependencies is crucial for accurate predictions. Figure 8 illustrates an example of the intermediate attention fusion mechanism.

### Late fusion

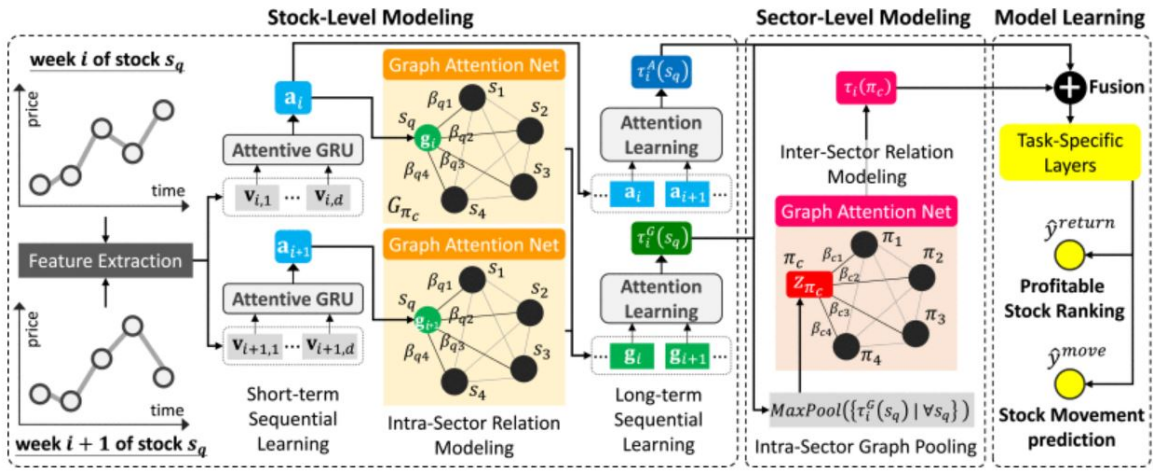


Figure 9: Example of late fusion from [46]. In this framework, the fusion between stock-level information and sector-level modelling is performed through the concatenation of feature embeddings, followed by a ReLU layer.

Late fusion, by contrast, delays the integration until after each modality has been processed by its own dedicated encoder. For instance, a graph neural network (GNN) might encode relational information between companies, while a Transformer processes news text, and a separate LSTM analyses historical prices. Each of these models produces an output embedding or prediction, and these outputs are then combined—typically via averaging, weighted sum, voting, or a shallow classifier—into the final decision. This approach is modular and flexible: each branch can be optimized separately and can use modality-specific architectures.

Despite the architectural diversity observed in multimodal financial forecasting models, the surveyed literature reveals a strong methodological convergence toward a small

set of underlying fusion principles. In particular, many architectures that differ in backbone design—ranging from recurrent networks and graph neural networks to attention-based and Transformer models—can be interpreted as implementing similar fusion logics. Early fusion approaches reflect the assumption that modalities are temporally aligned and jointly informative, enabling direct feature-level integration. Intermediate fusion mechanisms, including attention, gating, and message-passing schemes, operationalize a common principle of adaptive information selection, whereby the contribution of each modality is dynamically weighted based on contextual relevance. Late fusion architectures, often realized through ensembles or decision-level aggregation, embody the principle of conditional independence across modalities, prioritizing robustness and modularity over tight cross-modal coupling.

From this perspective, seemingly distinct architectures can be understood as different instantiations of the same fusion paradigms, tailored to specific data characteristics and modelling constraints. For example, attention-based fusion in sequence models and relation-aware message passing in graph-based models both serve to regulate cross-modal information flow under partial observability and modality imbalance. Making these connections explicit helps reconcile architectural heterogeneity with conceptual simplicity, and clarifies how recent multimodal models converge toward shared design principles rather than introducing fundamentally new fusion theories.

#### 2.4. Main Tasks

In financial forecasting, various modelling tasks can be defined based on what the system is trying to predict. These tasks are usually grouped into two main categories: *stock ranking* and *stock feature prediction*. Each task addresses a specific problem in financial decision-making and brings different advantages to investors, portfolio managers, and analysts.

##### *Stock Feature Prediction*

This category encompasses a range of tasks designed to predict specific quantitative indicators associated with individual stocks. It comprises several sub-tasks, each with different objectives and applications.

**Stock Movement Prediction.** This task is typically formulated as a classification

problem. The goal is to predict whether the price of a stock will go up, down, or remain stable over a certain period. This simplifies the modelling process and allows the use of standard classification models. Despite being relatively coarse, this task is highly actionable for short-term trading signals.

**Stock Return Prediction.** In this task, the model predicts the future return of an asset, defined as:  $r_{t+\Delta t} = \frac{P_{t+\Delta t} - P_t}{P_t}$ . This is typically approached using regression methods trained to minimize the expected squared error. The output is a continuous-valued forecast, and the task is sensitive to outliers and heavy tails in the distribution of returns.

**Volatility Prediction.** Volatility forecasting involves predicting the future variability of asset returns. Accurate volatility estimates are essential for derivative pricing, risk assessment, and hedging. Models may output point forecasts (e.g. standard deviation) or distributional estimates.

**Close Price Prediction.** This task aims to estimate the actual closing price of a stock for the next trading session or over a future horizon. It is a regression task and can provide useful benchmarks for stop-loss levels or price targets. Formally, we can express this as:  $\hat{P}_{t+1} = f_{\theta}(\mathcal{X}_t)$ , where  $\mathcal{X}_t$  represents features available at time  $t$ , and  $\hat{P}_{t+1}$  is the predicted closing price.

**Cross-Market Index Prediction.** Cross-market index prediction deals with forecasting the values of broader market indices, such as the S&P500 or Nasdaq. These indices reflect the aggregated performance of multiple assets and are influenced by microeconomic and macroeconomic variables. Formally, this can be written as:  $\hat{I}_{t+1} = f_{\theta}(S_t, M_t, T_t)$ , where  $S_t$  represents stock-level features,  $M_t$  macroeconomic indicators, and  $T_t$  textual or event-based data.

### *Stock Ranking*

Stock ranking aims to sort a list of stocks based on their expected performance, such as future return or risk-adjusted value. Unlike other tasks that try to predict exact prices or returns, stock ranking focuses on the relative order of the assets. This is especially useful in portfolio construction and decision support systems [45], where an investor might only be interested in the top-k performing stocks according to a model. In practice, this task assigns a score to each stock and produces a ranked list, enabling

strategies like long-short trading or factor investing without relying on precise value predictions. Formally, given a set of stocks  $\{s_1, s_2, \dots, s_n\}$  and a predictive model  $f_\theta$ , stock ranking seeks to compute a score  $r_i = f_\theta(s_i)$  for each stock such that the ranked list  $\{s_{(1)}, s_{(2)}, \dots, s_{(n)}\}$  satisfies:  $r_{(1)} \geq r_{(2)} \geq \dots \geq r_{(n)}$ .

### 3. A Taxonomy-Based Framework for Multimodal Stock Forecasting

To systematically organize this expanding body of literature, we propose a taxonomy-based framework that identifies and categorizes key dimensions along which existing works can be compared and analysed. This taxonomy serves both as a conceptual map and as a tool for highlighting current trends, gaps, and future research opportunities in the field.

Figure 10 categorizes existing multimodal forecasting studies along four orthogonal dimensions, which correspond to those introduced in section 2.

While the taxonomy is primarily organized along structural dimensions (input modalities, backbone architectures, fusion techniques, and tasks), it also provides a conceptual lens grounded in information fusion. In particular, different branches of the taxonomy reflect distinct fusion principles: hybrid and multimodal architectures emphasize modality complementarity, attention-based intermediate fusion supports relevance filtering and conflict resolution, and late fusion strategies implicitly address redundancy and robustness under partial observability.

The **Input-Modality** dimension specifies the data sources a model ingests. It spans: (i) market data such as OHLC price series and derived technical indicators; (ii) textual data drawn from social-media posts, official filings, and news articles; and (iii) relation data that encode firm-to-firm links—e.g., institutional ownership, governance structures, and industry or sector membership.

The **Backbone-Architecture** axis captures the neural family employed to process each modality. Most works rely on recurrent networks (LSTM/GRU) for temporal signals, graph neural networks (GCN/GAT and variants) for structured relational information, or hybrid stacks that combine multiple backbones to exploit complementary strengths.

The **Fusion-Technique** axis describes where and how modalities are integrated.

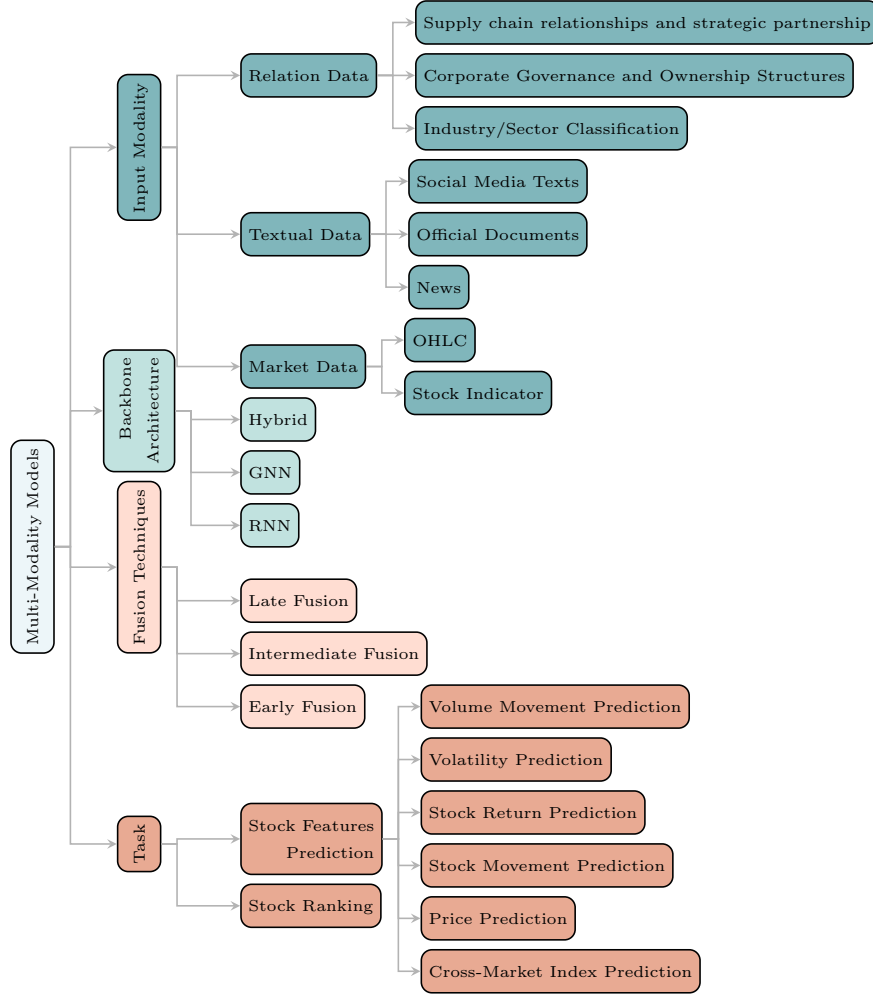


Figure 10: A taxonomy of Multi-Modality models for financial prediction, organizing input modalities, backbone architectures, fusion techniques, and tasks. Together with the accompanying discussion, the taxonomy provides a conceptual framework to interpret multimodal architectures in terms of information fusion principles such as modality complementarity, redundancy handling, and conflict resolution.

Early fusion concatenates features before encoding, intermediate fusion merges latent representations through cross-modal attention or gating, and late fusion aggregates modality-specific predictions at the decision layer. This categorization highlights the trade-off between mutual information exchange and architectural modularity.

Finally, the **Task** dimension groups downstream objectives. Under stock-features prediction we include fine-grained targets such as volume movement, volatility, returns, directional movement, price levels, and cross-market index values; a separate branch, stock ranking, focuses on ordering assets (e.g., top-K selection) rather than forecasting point estimates. Together, these four dimensions provide a compact yet comprehensive

map for positioning existing methods and identifying unexplored design combinations.

Beyond categorizing models by input modalities and architectural components, the proposed taxonomy also reflects implicit information fusion principles. In particular, different architectural patterns can be interpreted as mechanisms to exploit modality complementarity, handle redundancy across signals, and resolve conflicts arising from heterogeneous and partially inconsistent inputs.

The following sections apply this taxonomy to analyse representative works in multimodal financial forecasting, enabling a structured comparison of their design choices and methodological trends.

### *3.1. Input Modalities*

A central component in multimodal financial forecasting models is the selection and integration of diverse input modalities. Based on the literature, we identify three main categories: (i) market data, (ii) textual data, and (iii) relational or graph-based data.

**Market data.** Market data, primarily OHLCV time series enriched with technical indicators, serves as the foundational input for most quantitative forecasting models. For example, Wang et al. [52] propose a FFNN architecture enhanced by multimodal fusion strategies that include OHLC data. Similarly, Cheng et al. [19] combine market time series with graph-based relational structures to capture temporal and inter-asset dependencies. Another relevant work is Liu et al. [53], where the authors incorporate multi-scale temporal data along with market prices into a GNN-based framework.

**Textual data.** Textual data, including news, reports, and social media, provides additional signals on sentiment, macro trends, and firm-specific events. In Li et al. [54], the authors integrate online news with stock price data through a multimodal LSTM architecture that captures the impact of events. Sawhney et al. [55] uses both tweets and news to extract time-sensitive semantic features using attention mechanisms. Similarly, Hu et al. [56] focuses entirely on news-driven predictions, emphasizing the chaotic but informative nature of financial news. Another example is Ang et al. [57], where the model integrates news at both local and global scales to jointly forecast multiple financial indicators.

**Relational data.** The third category comprises relational or graph-based data, capturing structured relationships among financial entities through static or dynamic graphs such as co-movement networks, knowledge graphs, or industry hierarchies. In Qian et al. [58], a multi-relational GNN captures heterogeneous relations (e.g., sector, supply chain, ownership) and temporal evolution. Similarly, Chang et al. [19] highlights how price time series can be enriched with topological relationships via GNN layers.

### 3.2. Model Architectures

The backbone architecture of a multimodal financial forecasting model plays a pivotal role in determining how effectively it can process and integrate different input modalities. Across the literature, we identify five main architectural categories: recurrent networks (LSTM/GRU), graph neural networks (GNNs), and hybrid architectures that combine multiple of the above.

**RNN.** Farimani et al. [59] propose an approach that aims to predict the absolute exchange rate price of an asset at the next trading hour by leveraging multimodal data: news content, aggregated mood sentiment, and market technical indicators. The model builds a latent concept space for news by clustering word embeddings and vectorizing news documents based on term frequencies of these concepts. Sentiment scores are derived from a fine-tuned financial transformer model (FinBERT [60]), while market data include common technical indicators normalized adaptively to adjust to non-stationary distributions. Feature extraction involves applying LSTM layers on news data, and recurrent layers on mood and market time series to capture temporal dependencies. An attention mechanism dynamically weighs the importance of news, mood, and market features over time. Finally, the fused features are passed through dense layers to predict a price change offset, which is added to the lagged average closing price to generate the final price prediction. The model is trained end-to-end using MSE loss optimized by Adam.

In Liu et al. [45], the authors propose a model for stock ranking that leverages both textual data and historical price information. The two modalities are processed independently: textual inputs are first embedded using BERT and then further refined through LDA-based feature extraction, while historical price data are directly encoded.

Each modality is then passed through a separate LSTM network to capture temporal dynamics. Finally, the outputs are fused in a feed-forward layer and optimized using a pairwise ranking-aware loss function to produce a ranked list of stocks.

The framework proposed by Cheng et al. [61] predicts stock movements by integrating three types of information: technical indicators, news text, and inter-stock relationships. It first uses a tensor fusion module to combine technical and textual signals, capturing both their individual and joint effects. The resulting fused vectors, representing daily firm-level information, are processed by a GRU to model temporal dynamics and generate sequential embeddings. To capture the influence between firms—referred to as momentum spillover—the model constructs relational embeddings using a dynamic graph at each time step based on the correlations among stocks. In this way, the model dynamically learns the strength of inter-firm relations based on market signals, guided by attention mechanisms that account for both node attributes and learned graph structures. Finally, sequential and relational embeddings are concatenated and passed to a prediction module to classify the future stock movement direction. The entire model is trained end-to-end using a cross-entropy loss.

Zao et al. [62] propose a model that predicts stock movements by learning sequential embeddings through a GRU that processes fused multimodal market signals, combining technical indicators and news sentiment over time. This captures the temporal dynamics of each stock. Relational embeddings are derived from a bi-typed market knowledge graph linking companies and executives, using a Dual Attention Network to model both inter-class and intra-class interactions via hierarchical attention mechanisms. The sequential embeddings from the GRU and the relational embeddings are then combined and passed through a feed-forward network with softmax for prediction. The entire system is trained end-to-end with cross-entropy loss.

The PEN architecture [63] primarily relies on recurrent neural networks, specifically LSTM and GRU. It uses a bidirectional GRU layer to generate low-dimensional text embeddings from daily news. A Text Memory Unit, inspired by LSTM, preserves important textual information over time. Text and price data are combined through recurrent modules to learn shared representations. A recurrent variational autoencoder based on GRU is then used to generate stock movement predictions. Finally, a temporal

attention mechanism aggregates predictions across time steps for improved accuracy. Overall, PEN focuses on LSTM and GRU to capture temporal dynamics in both text and price data, without employing graph neural networks.

**GNN.** Pei et al. [64] propose a framework that jointly captures temporal evolution and static relationships among stocks using a dual-graph approach. The model constructs two dynamic graphs: a temporal graph that changes over time to reflect evolving market behaviour, and a static relational graph capturing long-term dependencies. To extract representations, the authors employ a GCN with contrastive learning to align node embeddings across time steps, encouraging consistency while allowing for temporal variation. This dual perspective enhances the model’s ability to represent both persistent structures and dynamic patterns in financial data.

Kumar et al. [65] present a model that uses dynamic graphs to reflect the time-varying relationships among financial assets. The architecture incorporates temporal snapshots of asset interactions, which are fed into a GAT to model both the evolution of the graph and the time series. The GAT module learns context-aware features from each temporal graph, effectively capturing volatility-relevant patterns across changing market structures.

In Sawhney et al. [66], the authors propose a framework for stock movement prediction that integrates both tweets and historical price data. After encoding the two modalities into feature vectors, these are used as node attributes in a Graph Attention Network (GAT), where each node represents a stock. The GAT is then trained to perform node classification, predicting the binary movement (UP or DOWN) of each stock based on its features and its relationships with other stocks.

In Xu et al. [67], the authors present a GCN-based approach for stock movement prediction that leverages event information related to individual stocks. The core intuition is that financial events can significantly influence stock behaviour, and modelling their relational structure can enhance predictive performance. The model uses a Graph Convolutional Network (GCN) to capture the importance of each event by learning from the event-stock interaction graph. For each stock, the representations of all associated events are concatenated, and the resulting vector is passed through a dense layer to generate the final prediction.

Feng et al. [68] use LSTM networks combined with an attention mechanism to extract a feature matrix from various technical indicators (such as Exponential Moving Average, Average Price, etc.). This feature matrix is then used as the input representation for the nodes in a stock relation graph. A Graph Attention Network (GAT) is applied to this graph to capture inter-stock dependencies, and the final output is obtained through a prediction layer optimized using a point-wise regression loss objective.

Gao et al. [49] propose TRAN, a framework that combines market data with features extracted from text for stock ranking prediction. Specifically, market data are used as node features in a graph, while an attention mechanism is employed to integrate textual and numerical information. The resulting attention scores are used as edge weights to represent the strength of relationships between stocks. A Graph Convolutional Network (GCN) is then applied to this graph to generate final stock rankings.

Qian et al. [58] introduce a sophisticated GAN framework that integrates multiple types of financial relations (e.g., industry, supply chain, co-movement) into a dynamic graph structure. The model combines relational-aware graph attention with temporal encoding, allowing it to simultaneously process heterogeneous relations and their temporal evolution. The GAN architecture includes multi-relational attention layers that weigh the importance of each edge type and time step, enabling a comprehensive and dynamic understanding of stock interactions for investment prediction.

**Hybrid.** Hybrid architectures combining graph-based and sequence-based components can be interpreted as explicitly modelling modality complementarity, where relational dependencies captured by GNNs are integrated with temporal and textual dynamics learned by sequential or attention-based encoders.

Hsu et al. [46] introduce a hybrid GRU-GAT framework for ranking stocks based on their predicted return ratios and finding future stocks with positive movements. Starting from historical price data, GRU layers are used to extract temporal features, which are then incorporated into stock-level and sector-level interaction graphs. These graphs are encoded using multi-head attention layers, and the resulting node representations are optimized via task-specific loss functions designed to guide stock ranking and identify assets with positive future movements.

Zhao et al. [69] proposed a hybrid model that combines GCN and LSTM to capture

both relational and temporal patterns in financial data. It integrates long-term correlations between stocks through a multi-relational stock graph built from historical price and volume data. For short-term modelling, separate GCNs are applied to price and volume graphs, followed by a graph-aware LSTM that captures temporal dependencies within each stock. News events are encoded via an attention-based LSTM and fused with stock features at the final time step, enriching the representation. Additionally, it uses a Siamese network to project price and volume representations into a shared space and optimizes their correlation. The final prediction is made through a fully connected layer.

Cheng et al. [38] propose a hybrid architecture that combines GCN and LSTM-based models to learn rich event embeddings for quantitative investment. It integrates three components: (1) a relation extraction module using BiLSTM with attention to detect entity relations from financial news; (2) an event representation module that uses a Neural Tensor Network (NTN) and GCNs over a financial knowledge graph (FinKG) to capture structured and contextual information; and (3) a multi-source attention mechanism to fuse embeddings from events, relations, and the graph. This GCN-LSTM hybrid framework captures both direct and lead-lag effects of events on stock movements.

Lazcano et al. [32] propose an architecture for time series prediction that is a hybrid model called BiLSTM-GCN that combines two neural network types to leverage their complementary strengths. The core architecture is a Graph Convolutional Network combined with LSTM (GCN-LSTM). This model constructs a graph from the correlation matrix of the time series data, enabling it to capture spatial relationships among nodes through graph convolutional layers. The spatial features extracted by the GCN are then processed by LSTM layers to model temporal dynamics. Dropout and dense layers are added to improve generalization and performance. This hybrid architecture, named BiLSTM-GCN, effectively integrates the temporal modelling capabilities of the BiLSTM with the spatial-temporal learning of the GCN-LSTM, leading to improved prediction accuracy.

The architecture proposed in Li et al. [70] is a hybrid model combining LSTM and GCN to predict overnight stock movements. News headlines are encoded with

LSTM and attention, merged with stock embeddings, and used as node features in a stock correlation graph. The graph uses RGCN layers to capture positive and negative stock relationships. LSTM units between GCN layers help avoid over-smoothing by dynamically gating information. A global node models overall market trends. The final node representations are used for binary stock movement prediction.

The ST-Trader model [71] tackles stock price prediction by capturing both spatial and temporal dependencies. First, a Variational Autoencoder (VAE) is used to encode static fundamental features of each stock into a low-dimensional latent space. Based on the distances between these latent vectors, a spatial graph is constructed, where nodes represent stocks and edges reflect their learned similarity. To model temporal dynamics, the framework employs a hybrid GCN-LSTM architecture: Graph Convolutional Networks apply spectral filters (via Chebyshev polynomials) to propagate information across the graph, while LSTM units capture sequential patterns in historical stock prices. By integrating graph-based spatial structure with time-series modelling, ST-Trader generates more informed predictions of future stock prices.

Liu et al. [72] propose a hybrid architecture that combines multiple deep learning techniques to integrate market, text, and social data for stock return prediction. It uses LSTM for time series market data, BERT for encoding financial text, and graph neural networks (GNN) to capture social network relationships. These diverse feature representations are fused through a multimodal attention mechanism, allowing the model to adaptively weigh each data source. Finally, classic machine learning methods like XGBoost and LASSO select the most predictive factors. This hybrid approach effectively leverages different data types and modelling techniques to improve factor robustness and prediction accuracy.

This diversity of architectures reflects the multifaceted nature of financial forecasting tasks and the need for models that can flexibly adapt to heterogeneous data sources. The ongoing shift toward transformer and graph-based models, and especially their hybrid combinations, signals a promising direction for future work.

### *3.3. Fusion Approach*

**Early Fusion.** Early fusion techniques combine data from multiple sources at the input level, before being processed by the model. This approach concatenates or oth-

erwise merges raw or low-level features (e.g., prices, volumes, textual indicators) into a single input vector that is fed into the model. The advantage is that the model learns from all modalities simultaneously from the very beginning, potentially capturing simple correlations across data types.

Cheng et al. [61] propose an early fusion strategy using a Tensor Fusion Module to jointly model numerical and textual features for stock prediction. For each stock, technical indicators and textual embeddings (e.g., from news articles) are first extracted and then fused through a bilinear tensor product that captures cross-modal interactions. Additionally, a linear transformation of the concatenated features preserves their independent contributions. The final fused representation is obtained by combining both terms and applying a non-linear activation function. This approach enables the model to learn task-specific and stock-specific feature interactions in a unified representation.

In Li et al. [70], the authors construct a stock correlation graph where each node represents a stock and edges reflect positive or negative correlations based on historical prices. Each node is enriched with news text, encoded via an LSTM followed by an attention mechanism that uses the stock embedding as a query. The resulting text representation is concatenated with the stock embedding to form the node features. This multimodal graph, combining market and textual signals, is then processed by a hybrid LSTM-RGCN model to predict stock movements.

Deng et al. [48] propose an early fusion approach to represent structured event information extracted from financial news. The method starts by applying Open Information Extraction (OpenIE) [73] to transform unstructured news text into structured event tuples, each consisting of a subject, predicate, and object. These components are then linked to a Knowledge Graph to enrich them with semantic and relational information. To capture broader contextual knowledge, the model also includes the immediate neighbours of the linked entities in the graph. For each element of the event tuple, three types of features are extracted: embeddings from the Knowledge Graph, contextual embeddings from neighbouring entities, and standard word embeddings. All these representations are then concatenated into a single vector. This fused representation is used as input for the stock movement prediction tasks.

Xu et al. [67] propose a multi-step fusion mechanism that integrates event text

and stock-specific historical context. First, events extracted from financial news are encoded using a type-specific multi-head attention mechanism, which highlights the most relevant tokens based on the event type. These daily event embeddings are then processed by an LSTM to capture temporal dependencies over recent days. In parallel, the model encodes each stock’s historical context by processing past events and their market feedbacks (e.g., price or volume changes) with two LSTMs, whose outputs are concatenated. To model the stock-dependent effect of event information, the event embedding and the stock context embedding are fused via a feed-forward network. This produces a stock-specific modulation of the event signal, enabling the model to capture how different stocks react differently to similar events.

Zhao et al. [62] present a model that performs early fusion of multi-modal time-series signals, specifically combining technical indicators and sentiment features derived from news. The authors note that stock prices are influenced by market signals over multiple days, so their model takes into account the historical sequence of fused features when predicting future movements. The fusion mechanism is based on a Neural Tensor Network, which captures the complex interactions between the two modalities—technical and sentiment—by modelling their relationships across multiple dimensions. The resulting fused representation for each day is then passed through a GRU layer, and the final hidden state is used to encode the temporal dynamics of each stock.

MAN-SF: Multipronged Attention Network for Stock Forecasting, proposed by Sawhney et al. [66], combines historical price data and social media signals using a generalized bilinear fusion mechanism, which explicitly models pairwise interactions between the two modalities. Price features are extracted using a GRU followed by temporal attention, which highlights the most informative trading days. Social media information is encoded hierarchically: a GRU with intra-day attention identifies impactful tweets within each day, and a second GRU with temporal attention aggregates information across days. The resulting price and tweet representations are then fused through a bilinear transformation, which captures their joint dynamics more effectively than simple concatenation or averaging. This fusion layer enables the model to learn richer cross-modal interactions, leading to better market trend prediction.

**Interpretability considerations.** From an interpretability perspective, early fusion offers limited transparency at the modality level. While feature attribution methods ([74]) can be applied to the fused input, the joint transformation of heterogeneous signals often leads to strong representational entanglement, making it difficult to disentangle the contribution of individual modalities (e.g., market data versus textual signals). As a result, explanations tend to remain at a low-level feature granularity ([75]) and may lack clear semantic correspondence with domain concepts, which can hinder auditability in high-stakes financial applications.

**Intermediate Fusion.** Intermediate fusion techniques integrate data from multiple sources within the model during training. Unlike early fusion, which combines raw inputs before processing, or late fusion, which merges separate model outputs, intermediate fusion merges features at hidden layers. This allows the model to learn joint representations that capture complex interactions between different data modalities, improving performance on tasks where multiple types of information must be considered together to produce a single final output.

DGRCL proposed in Pei et al. [64] uses an intermediate fusion strategy to combine temporal stock data and structural company relations. It first constructs a dynamic graph at each time step by computing pairwise stock similarities using Dynamic Time Warping (DTW) [76]. Then, it generates initial node embeddings that capture both stock-specific temporal trends and relational constraints derived from company links. These enhanced embeddings are passed through an RNN to capture sequential dependencies and produce final node representations. The fusion of modalities occurs within the RNN, allowing the model to jointly learn from both temporal patterns and structural relations before making predictions.

The Melody-GCN model [53] combines stock price data and media text information through an intermediate fusion strategy within its processing pipeline. Initially, the model separately analyzes historical price movements and text embeds derived from media sources, extracting features from each modality along multiple temporal scales. Crucially, it does not merge these two data types at the input or output stages. Instead, it fuses them during feature extraction: the encoded price and text representations are combined midway, processed jointly, and then used to enhance each other. This

allows price-based insights to inform text understanding, and vice versa, creating richer, context-aware representations.

In the Dandelion model [77], intermediate fusion occurs after each modality (such as time series, fundamentals, and graphs) is processed by a dedicated encoder. The resulting latent representations are then integrated through a multimodal attention mechanism, which dynamically weighs the contribution of each modality based on its relevance to the task. To encourage alignment across modalities, a regularization term is applied to minimize their distance in the shared latent space. This fusion step takes place before prediction and enables selective information sharing across related tasks.

Hu et al. [56] propose a stock trend prediction model that performs intermediate fusion of textual data. Individual news articles for each day are first converted into vectors and weighted through an attention mechanism to highlight the most relevant ones. These daily aggregated vectors are then combined along the temporal sequence using another attention layer that selects the most influential days. Thus, fusion occurs in two stages—at the news level and at the temporal level—gradually integrating textual information into the final representation used for prediction.

Prediction-Explanation Network (PEN) for stock movement prediction, proposed by Li et al. [63], performs intermediate fusion of text and price data through its Shared Representation Learning (SRL) module. SRL integrates daily text embeddings and historical price features by first selecting relevant texts via a Text Selection Unit (TSU) that assigns importance scores to each text. Then, a Text Memory Unit (TMU), inspired by LSTM, preserves important textual information over time. Finally, the Information Fusion Unit (IFU) merges the text memory and price data into a shared hidden representation, which captures their interaction and is recurrently updated across time steps. This fused representation is used downstream for variational inference and prediction, enabling the model to jointly leverage textual and price signals in a temporally-aware manner.

Shi et al. [78] propose a model that leverages four types of graphs representing different stock relations (industry/region, concept, volatility) and uses GCN to extract temporal embeddings that capture both relational and time-varying information. These embeddings are then fed into an LSTM to predict stock price movements. To

enhance prediction performance, embeddings from the different graphs are fused using three strategies: concatenation, mean-filtering, and max-filtering. These fusion methods enable efficient integration of multiple knowledge sources, balancing between increased embedding dimensionality (concatenation) and information compression (filtering), thereby improving the model’s predictive power.

The model proposed in Feng et al. [68] fuses temporal and spatial financial data using an intermediate fusion mechanism within a unified architecture. It processes stock time series with an attention-based LSTM to capture key temporal features and extracts spatial relationships from stock correlations via a stacked graph neural network combining graph convolution and graph attention layers. These two feature sets—temporal embeddings and graph-based spatial embeddings—are integrated during training through joint optimization, allowing the model to learn complex interactions between stock trends over time and their inter-stock correlations for improved stock ranking predictions.

**Interpretability considerations.** Intermediate fusion strategies provide a partial improvement in interpretability by preserving modality-specific representations while enabling controlled cross-modal interaction. In particular, attention or gating-based mechanisms can expose modality and time-specific relevance signals, allowing practitioners to inspect how information from one modality influences another during representation learning ([79]). However, this interpretability remains partial: while such mechanisms can support qualitative tracing of cross-modal influence, they do not necessarily guarantee faithful or causal explanations, especially when attention weights are used as a proxy for importance ([80]).

**Late Fusion.** Late fusion techniques combine the outputs of separate models, each trained on a different data modality. In this approach, each model processes its input independently and produces its own prediction or feature representation. These outputs are then aggregated—using methods like averaging, weighted voting, or a meta-learner—to produce the final prediction. Late fusion is particularly useful when modalities are highly heterogeneous or asynchronously available, and it allows leveraging specialized models for each data source.

Liu et al. [72] adopts a late fusion approach to predict stock returns by first in-

independently learning feature representations from multiple modalities—market data (via LSTM/Transformer), textual data (via BERT), and social data (via GNN). Each modality is processed by a dedicated model, and their outputs are later combined using a multimodal attention mechanism, which assigns adaptive weights to each modality. The resulting fused representation is then passed to a factor selection module (e.g., XGBoost or LASSO) to extract the most predictive features.

The fusion mechanism in Lazcano et al. [32] follows a late fusion approach, where two separate models—a BiLSTM and a GCN-LSTM—are first pretrained independently on the time series data. Their outputs are then concatenated to form a combined feature vector, which is passed through additional dense layers to integrate the complementary information extracted by each model. This approach allows the hybrid network to leverage both temporal dependencies captured by the BiLSTM and spatial-temporal correlations modelled by the GCN-LSTM, resulting in improved prediction accuracy.

The model proposed in Xiang et al. [81] employs a late fusion strategy to combine temporal and relational information. Specifically, it first encodes each stock’s historical price sequence and its graph-based neighbourhood separately. Temporal representations are obtained using a transformer, while relational features are extracted through a heterogeneous graph attention mechanism over dynamic correlation-based graphs. Late fusion occurs after these independent encodings: an attention mechanism integrates the stock’s own features with those from positively and negatively correlated neighbours. This late-stage combination allows the model to preserve the specific strengths of temporal and structural signals before merging them into a unified representation for prediction.

In Sawhney et al. [82], the fusion of different data sources is achieved through an ensemble approach that combines outputs from separate models trained on distinct data types: the textual analysis of earnings calls and the historical stock price data. Historical stock prices over the 30 days before an earnings call are modelled using Support Vector methods—Support Vector Regression for volatility and Support Vector Classification for price movement. These predictions are then combined with outputs from models processing textual and aligned speech features from the earnings calls. To optimize this fusion, the ensemble weights assigned to each component are carefully

tuned by experimenting with different values in small increments, allowing the model to find the best balance between market data and earnings call information for improved prediction performance.

Zao et al. [69] predict trading volume movement by combining long-term stock relations, short-term price and volume fluctuations, and sudden financial news. It builds a graph connecting stocks based on historical price and volume correlations, capturing different types of relationships. Price and volume data are processed separately through graph neural networks and a specialized LSTM to model temporal changes while reducing noise. Sudden events from news headlines are encoded and fused by adding them as the last time step in the sequence, creating a combined temporal graph that integrates news with transaction data. The fusion happens by merging stock embeddings, encoded news features, and processed price and volume vectors into a single representation used for final prediction. This approach allows the model to effectively combine diverse data sources for better volume movement forecasting.

In RGStockNet [83], the fusion step integrates two complementary representations to improve stock trend prediction. The first is the relational embedding, which captures structural knowledge about inter-stock relationships learned from a knowledge graph built on pairwise time series interactions. This embedding provides a global market context for each company. The second is the time series embedding, obtained from a dedicated encoder that processes the recent historical data of the target stock, such as daily prices and other numerical indicators. This representation reflects the local temporal dynamics of the individual stock. Fusion is performed by a fully connected network which outputs the probability distribution over the possible future trends.

To summarize, while early, intermediate, and late fusion strategies are all widely adopted in multimodal financial forecasting, the reviewed literature suggests that their effectiveness is strongly dependent on the characteristics of the data and the target setting, rather than on a universally superior fusion paradigm. As emerges from the analysis of the surveyed works, early fusion is typically employed when input modalities are temporally aligned and relatively homogeneous, such as technical indicators and sentiment scores aggregated at the same daily frequency. Approaches based on early fusion

(e.g., [62, 48, 61, 66]) benefit from simplicity and tight cross-modal interaction at the feature level, but they implicitly assume consistent availability and comparable noise levels across modalities. As reported in several studies, this assumption may limit robustness in realistic financial scenarios, where textual signals are sparse, asynchronous, or noisy, and where missing modalities can significantly degrade performance.

In contrast, intermediate fusion strategies—often implemented through attention or gating mechanisms—have become increasingly prevalent in recent works (e.g., [64, 53, 56, 63, 77]). These approaches explicitly model cross-modal dependencies at the representation level, allowing the model to dynamically weight modalities based on their relevance. From the surveyed literature, intermediate fusion appears particularly well suited to financial environments characterized by modality imbalance or partial observability, such as settings where news coverage is uneven across stocks or time. Moreover, attention-based fusion enhances interpretability by exposing modality-level importance scores. This design choice reflects a broader trend toward adaptive and context-aware fusion mechanisms, which trade architectural simplicity for improved flexibility under heterogeneous data conditions.

Late fusion approaches, on the other hand, are primarily adopted in scenarios where modalities are highly heterogeneous or weakly aligned, including cross-market and multi-source settings (e.g., [72, 32, 81, 82]). By processing each modality independently and combining predictions at the decision level, late fusion offers modularity and robustness to missing inputs, as individual components can operate even when one modality is unavailable. However, as noted in the reviewed studies, this modularity comes at the cost of reduced cross-modal interaction, potentially limiting the model’s ability to capture fine-grained dependencies between textual, temporal, and relational signals. As a result, late fusion is often preferred in practical deployments emphasizing stability and scalability over maximal predictive synergy.

**Interpretability considerations.** Late fusion is generally the most interpretable fusion strategy at the modality level, as it preserves separate processing pipelines for each data source ([84]) and combines them only at the decision stage. This architectural separation allows predictions to be decomposed by modality, facilitating accountability, auditing, and error analysis. For this reason, late fusion is often preferred in regulated

or risk-sensitive financial settings ([85]), where understanding and documenting the contribution of each information source is as important as predictive performance.

### 3.4. Predictive tasks

**Stock movement prediction.** Stock movement prediction focuses on the directional change in the price of a stock over a given time horizon. Many studies formalise the task as a classification (binary or multi-class) to identify whether a stock will go up, down or remain stable. For example, in Li et al. [54], a fusion of financial news and market data is used to predict binary price movements. Similarly, Xiang et al. [81] integrate temporal market data and graph structures to improve the performance of binary classification. Other works such as Pei et al. [64], Cheng et al. [38] and Ye et al. [86] similarly frame the forecasting task as binary classification, exploiting graph representations and multimodal inputs to capture complex dependencies.

In Cheng et al. [19], the assignment of labels is determined using two thresholds: an upper threshold  $r_{up}$  and a lower threshold  $r_{down}$ . If the return  $R_r$  is greater than or equal to  $r_{up}$ , the movement is classified as UP; if it is less than or equal to  $r_{down}$ , it is classified as DOWN. When the return falls between these two thresholds, it is considered NEUTRAL.

This task remains a dominant choice in the literature due to its interpretability, ease of labelling, and practical relevance in algorithmic trading applications.

**Stock return.** This task aims to estimate the percentage change in a stock’s price over a specified time period, providing insight into potential future gains or losses. It is often modelled as a regression problem. In Wang et al. [52], by introducing adaptive feature selection modules, multimodal fusion modules, and regularization strategies, the proposed model can dynamically adjust feature weights, integrate multiple modality information (such as historical prices, technical indicators, and market sentiment), and effectively suppress overfitting problems. Zuou et al. [77] propose a time series forecasting framework named Dandelion, which leverages multiple modalities for the prediction.

**Cross-market index prediction.** It addresses the forecasting of foreign or international stock indices by leveraging multimodal signals. Lee et al. [87] exemplifies this

strategy. The authors developed stock prediction models that combine information from the South Korean and US stock markets by using multimodal deep learning.

**Close price prediction.** The objective of close price prediction is to forecast the actual future price of a stock rather than its return. Farimani et al. [59] employ a hybrid model for price regression that combines different components (FinBERT, CNN and LSTM).

**Volatility prediction.** It aims to estimate the expected variability of a stock’s returns, a critical component for risk-sensitive applications. For example, Ibrahim et al. [88] introduced the model DynGWN that relies on a Tensor Graph Convolutional Module (TGCM), which captures dynamic trends in graphs effectively in the time-varying graph representations.

**Multi-task learning.** A single model is trained to predict multiple financial variables simultaneously, such as stock movement, volatility, or return. This approach is motivated by the fact that these tasks are often correlated: for example, volatility patterns can provide valuable context for predicting stock direction or return. Architecturally, multi-task models typically share the same feature extraction and fusion layers but add task-specific output heads, each corresponding to a distinct predictive objective. In Sawhney et al. [82], the authors propose a multitask approach that uses textual features and attentive audio alignment to predict the movement and volatility of the stock price simultaneously. Similarly, Ang et al. [57] propose Guided Attention Multimodal Multi-task Network (GAME), a model that captures both global and local multimodality information for return and volatility predictions.

**Stock ranking.** In financial forecasting, stock ranking is a task that goes beyond binary or multi-class classification by aiming to order a set of stocks based on a specific target signal, such as expected return, profitability, or investment potential. Rather than predicting individual labels for each stock, the goal is to generate a relative ranking that reflects the most promising investment opportunities. This setup is particularly useful in portfolio construction or recommendation systems, where the interest lies in

selecting the top-K performing stocks rather than predicting the behaviour of each in isolation.

Liu et al. [45] focuses on exploiting the relationships among stocks through multi-modal data and graph-based learning methods to solve the stock rank prediction. They propose a Multi-modal Temporal Dynamic Graph method (MTDGraph) with the point-wise regression loss and ranking-aware loss to obtain the appropriate stock rank list.

Sawhney et al. [55] propose a hierarchical approach to learning-to-rank that uses textual data to make time-sensitive predictions to rank stocks based on expected profit.

In Liu et al. [45], textual features extracted from documents using LDA are combined with historical market data. These two modalities are processed independently and then fused in a fully connected layer to predict a ranking of stocks.

Lastly, Gao et al. [49] constructs a graph of the stock relation and extracts features from stocks and documents for the recommendation of the stock according to the ranking of the return ratio.

Table 1 summarizes a collection of works in multimodal financial forecasting, comparing the input modalities used, model architectures, fusion strategies, and prediction tasks. In this context, the market modality refers to all numerical features directly related to the stock, in addition to the target variable (e.g., future return or price movement). Typical examples include historical prices, technical indicators (such as moving averages or RSI), trading volume, and volatility measures.

## 4. Evaluation and Comparative Analysis

### 4.1. Data source and Datasets

**Market data.** Most of the works reviewed in this survey rely on financial market data obtained from Yahoo Finance<sup>6</sup>, a widely used open-access platform that provides historical and real-time data on stock prices, volumes, indices, and fundamental indicators (e.g., P/E ratios, dividends, earnings). The main strengths of Yahoo Finance include its accessibility, the breadth of data coverage, and the simplicity of use through both its web interface and APIs (such as `yfinance`<sup>7</sup> in Python). However, it has some

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<sup>6</sup><https://finance.yahoo.com/>

<sup>7</sup><https://pypi.org/project/yfinance/>

Table 1: Overview of analysis dimensions, detailing input modalities, model architectures, fusion strategies (early  $\oplus$ , intermediate  $\odot$ , late  $\cup$ ), and prediction tasks (cross-market index prediction  $\mathbb{G}$ , price prediction  $\$$ , stock movement prediction  $\mathbb{L}$ , stock ranking  $\mathbb{I}$ , stock return prediction  $\%$ , volatility prediction  $\text{⚡}$ , volume movement prediction  $\mathbb{L}$ ).

| Paper                | Input modality |         |          | Architecture | Fusion approach     | Task                  |
|----------------------|----------------|---------|----------|--------------|---------------------|-----------------------|
|                      | Market         | Textual | Relation |              |                     |                       |
| Li et al. [54]       | ✓              | ✓       |          | RNN          | $\oplus$            | $\mathbb{L}$          |
| Liu et al. [72]      | ✓              | ✓       | ✓        | Hybrid       | $\cup$              | $\%$                  |
| Lee et al. [87]      | ✓              |         |          | RNN          | $\oplus \odot \cup$ | $\mathbb{G}$          |
| Farinami et al. [59] | ✓              | ✓       |          | RNN          | $\cup$              | $\$$                  |
| Wang et al. [52]     | ✓              |         | ✓        | FNN          | $\cup$              | $\%$                  |
| Cheng et al. [19]    | ✓              | ✓       | ✓        | GNN          | $\oplus$            | $\mathbb{L}$          |
| Zhou et al. [77]     | ✓              | ✓       | ✓        | Hybrid       | $\odot$             | $\%$                  |
| Xiang et al. [81]    |                |         | ✓        | GNN          | $\cup$              | $\mathbb{L}$          |
| Ibrahim et al. [88]  |                |         | ✓        | Hybrid       | $\cup$              | $\text{⚡}$            |
| Pei et al. [64]      | ✓              |         | ✓        | GNN          | $\odot$             | $\mathbb{L}$          |
| Kumar et al. [65]    |                |         | ✓        | GNN          | $\odot$             | $\text{⚡}$            |
| Hsu et al. [46]      |                |         | ✓        | Hybrid       | $\cup$              | $\% \mathbb{L}$       |
| Lazcano et al. [32]  |                |         | ✓        | Hybrid       | $\cup$              | $\$$                  |
| Ang et al. [57]      |                | ✓       | ✓        | GNN          | $\cup$              | $\mathbb{L} \text{⚡}$ |
| Sawhney et al. [82]  |                | ✓       |          | RNN          | $\cup$              | $\mathbb{L} \text{⚡}$ |
| Liu et al. [53]      | ✓              | ✓       |          | GNN          | $\odot$             | $\mathbb{L}$          |
| Zhao et al. [69]     |                | ✓       | ✓        | Hybrid       | $\cup$              | $\mathbb{I}$          |
| Ye et al. [86]       | ✓              |         | ✓        | Hybrid       | $\odot$             | $\mathbb{L}$          |
| Liu et al. [45]      |                | ✓       | ✓        | RNN          | $\cup$              | $\mathbb{I}$          |
| Cheng et al. [38]    | ✓              | ✓       | ✓        | Hybrid       | $\odot$             | $\mathbb{L}$          |
| Cheng et al. [61]    |                | ✓       | ✓        | RNN          | $\oplus$            | $\mathbb{L}$          |
| Sawhney et al. [66]  |                | ✓       |          | GNN          | $\oplus$            | $\mathbb{L}$          |
| Li et al. [70]       |                | ✓       |          | Hybrid       | $\oplus$            | $\mathbb{L}$          |
| Hu et al. [56]       | ✓              | ✓       |          | RNN          | $\odot$             | $\mathbb{L}$          |
| Sawhney et al. [55]  |                | ✓       |          | RNN          | $\cup$              | $\mathbb{I}$          |
| Li et al. [63]       |                | ✓       |          | RNN          | $\odot$             | $\mathbb{L}$          |
| Deng et al. [48]     |                | ✓       | ✓        | RNN          | $\oplus$            | $\mathbb{L}$          |
| Qian et al. [58]     | ✓              |         | ✓        | GNN          | $\odot$             | $\mathbb{L}$          |
| Shi et al. [78]      | ✓              |         | ✓        | Hybrid       | $\odot$             | $\mathbb{L}$          |
| Li et al. [83]       | ✓              |         | ✓        | Hybrid       | $\cup$              | $\mathbb{L}$          |
| Zhao et al. [62]     | ✓              | ✓       |          | GRU          | $\oplus$            | $\mathbb{L}$          |
| Hou et al. [71]      |                |         | ✓        | Hybrid       | $\oplus$            | $\mathbb{L}$          |
| Xu et al. [67]       | ✓              |         | ✓        | GNN          | $\oplus$            | $\mathbb{L}$          |
| Feng et al. [68]     | ✓              |         | ✓        | GNN          | $\odot$             | $\mathbb{I}$          |
| Gao et al. [49]      | ✓              | ✓       | ✓        | GNN          | $\odot$             | $\mathbb{I}$          |

limitations: the data may be subject to occasional inconsistencies or missing entries, especially in high-frequency settings or for less-liquid assets.

**Textual Information.** Textual data employed in multimodal financial prediction tasks are typically sourced either from proprietary platforms or publicly available datasets. A substantial body of research relies on financial news articles and reports disseminated by major media outlets such as *Reuters*<sup>8</sup>, *Bloomberg*<sup>9</sup>, and *MarketWatch*<sup>10</sup>. These sources offer high-quality, timely, and domain-specific content that is instrumental in capturing market sentiment and detecting event-driven signals. In certain instances, access to such data is facilitated through institutional partnerships or previously compiled corpora—for example, the dataset introduced by Duang et al. [89], which includes Reuters and Bloomberg articles collected between October 2006 and December 2015. In other instances, publicly available datasets are utilized to promote reproducibility and facilitate model comparability. The *CMIN-US* dataset [90] comprises extensive financial texts alongside stock price time series data from both the U.S. and Chinese markets. Another widely adopted resource is the *ACL18* dataset [91], which integrates two modalities: social media data and historical price information. It includes 387,045 tweets related to 70 companies spanning seven industries. The *2017 Earnings Conference Calls dataset* [92] provides transcripts of earnings calls sourced from *Seeking Alpha*<sup>11</sup>, annotated with metadata such as speaker identities and their corresponding speech content.

#### 4.2. Evaluation Metrics in Multimodal Financial Forecasting

To evaluate multimodal models in financial forecasting, studies employ diverse metrics spanning classification, regression, ranking, and financial performance indicators [19, 64, 53, 48]. In classification tasks, metrics such as accuracy, precision, recall, and F1-score are common, though class imbalance can distort results. The Matthews Correlation Coefficient (MCC) [93] is therefore often preferred, as it balances true and

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<sup>8</sup><https://www.reuters.com/>

<sup>9</sup><https://www.bloomberg.com>

<sup>10</sup><https://www.marketwatch.com/>

<sup>11</sup><https://seekingalpha.com/>

Table 2: Overview of additional characteristics of the analysed papers, including performance metrics (e.g., Accuracy (A), Precision (P), Recall (R), F<sub>1</sub>-score (F<sub>1</sub>), Information Ratio (IR), and Investment Return Ratio (IRR)); dataset accessibility (private  vs. public ); country of origin (China , Japan , South Korea , Taiwan , U.S. , and multi-country datasets ); and whether the approach has been applied in real-world scenarios.

| Paper                | Metrics                                                                            |                                      |                                     |                                      |                                      |                                       |                                      |                                        |                                      |                                      |                                                |                                       |                                   |                                   | Dataset Access                    |                                   |                                   | Country                           | Real-world application            |                                   |                                   |
|----------------------|------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------|--------------------------------------|--------------------------------------|---------------------------------------|--------------------------------------|----------------------------------------|--------------------------------------|--------------------------------------|------------------------------------------------|---------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
|                      |                                                                                    |                                      |                                     |                                      |                                      |                                       |                                      |                                        |                                      |                                      |                                                |                                       |                                   |                                   | Market                            | Textual                           | Relation                          |                                   |                                   |                                   |                                   |
| Li et al. [54]       | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Liu et al. [72]      | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Lee et al. [87]      | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Farinami et al. [59] | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Wang et al. [52]     | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Cheng et al. [19]    | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Zhou et al. [77]     | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Xiang et al. [81]    | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Ibrahim et al. [88]  | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Pei et al. [64]      | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Kumar et al. [65]    | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Hsu et al. [46]      | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Lazcano et al. [32]  | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Ang et al. [57]      | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Sawhney et al. [82]  | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Liu et al. [53]      | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Zhao et al. [69]     | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Ye et al. [86]       | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Liu et al. [45]      | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Cheng et al. [38]    | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Cheng et al. [61]    | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Sawhney et al. [66]  | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Li et al. [70]       | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Hu et al. [56]       | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Sawhney et al. [55]  | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Li et al. [63]       | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Deng et al. [48]     | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Qian et al. [58]     | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> | <div><div></div><div>MCC</div></div> | <div><div></div><div>MdAPE</div></div> | <div><div></div><div>MRR</div></div> | <div><div></div><div>MSE</div></div> | <div><div></div><div>R<sup>2</sup></div></div> | <div><div></div><div>RMSE</div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> | <div><div></div><div></div></div> |
| Shi et al. [78]      | <div><div></div><div>A</div><div>P</div><div>R</div><div>F<sub>1</sub></div></div> | <div><div></div><div>AUC</div></div> | <div><div></div><div>IR</div></div> | <div><div></div><div>IRR</div></div> | <div><div></div><div>MAE</div></div> | <div><div></div><div>MAPE</div></div> |                                      |                                        |                                      |                                      |                                                |                                       |                                   |                                   |                                   |                                   |                                   |                                   |                                   |                                   |                                   |

false predictions and remains robust under imbalance. For regression tasks, Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE) are widely applied [72, 52, 59], with Median Absolute Percentage Error (MdAPE) offering resilience against outliers.

In stock ranking and portfolio allocation, ranking metrics are essential. Mean Reciprocal Rank (MRR@k) is used to evaluate the identification of top-performing assets [46, 45], complemented by Precision@k and Recall@k, which respectively assess accuracy and coverage within the top-k predictions. Together, these measures capture the prioritisation of profitable opportunities.

Overall, the choice of metrics reflects the heterogeneity of tasks, from movement classification to ranking and portfolio optimisation. While standard metrics provide comparability, specialised ones like MCC and MRR@k yield deeper insights into model behaviour and investment relevance, making metric selection critical to avoid misleading conclusions.

#### 4.3. *Analysed Markets*

Table 2 summarises the markets and indices analysed across the reviewed studies, a factor that critically shapes input data, prediction dynamics, and model generalisation. Most works focus on major benchmarks such as the S&P 500, widely used as a proxy for large-cap U.S. equities; models trained on this dataset (e.g. [55], [83], [66]) benefit from high liquidity, abundant textual data, and structured reporting. Others target Asian markets, notably the Chinese CSI 100 and CSI 300 indices (e.g. [19], [56], [62]), which differ from Western markets in investor composition, regulation, transparency, and government influence—factors that shape the relative value of data modalities. Some studies extend to cross-market or multi-region scenarios, which pose challenges such as unequal data availability, linguistic variation, and mismatched trading calendars. A model pre-trained on U.S. data, for example, may underperform in emerging markets without domain adaptation. These findings underscore the need for context-aware modelling: a modality or architecture effective in one market may not generalise to others, making market choice (see *Market* column) central to multimodal forecasting research.

## 5. Application in Real-World Scenarios

Several reviewed studies employ investment simulators to test the real-world effectiveness of their models, typically by applying trading strategies to historical market data and evaluating both absolute and risk-adjusted returns.

Li et al. [54] assess a tensor-based, multimodal LSTM analyser against eMAQT [94], TeSIA [95], AZFinText [96], and a top-N trading strategy. With a RMB 10,000 budget over three bearish months, their event-driven system achieved a 136.42% return versus 113.62% (TeSIA), 86.68% (eMAQT), and 70.30% (AZFinText), mainly through short-selling, with Sharpe ratio analysis confirming superior risk-adjusted performance.

Cheng et al. [19] report a real-world deployment of the MAGNN model, which integrates multimodal data into a portfolio management system. In a Spring Airlines case study, the model anticipated four major price swings, delivering over 60% excess profit and reducing maximum drawdown from -16.08% to -12.48%.

Similar to [19], the authors of [81] propose a portfolio management application supported by a desktop interface. For a given stock, the application visualises its historical market features, related stocks, and associated news articles, each annotated with buy (B) and sell (S) signals generated by the THGNN model.

In Li et al. [83], the authors deploy the RGStockNet framework in real-world trading scenarios, focusing on five companies, including Costco Wholesale Corp. (see Figure 11). They define a straightforward trading strategy: when the model predicts a Rise and no stock is currently held, a buy signal is triggered at the day’s closing price. Flat predictions are ignored, while a Drop prediction results in a sell signal at the next day’s closing price. Across these case studies, RGStockNet consistently outperforms baseline recurrent models. The authors note that standard architectures such as LSTM tend to produce repeated and static predictions over time, which impairs trading performance. RGStockNet mitigates this issue, thereby improving overall profitability.

Xu et al. [67] evaluate their REST framework through a simulated stock investment strategy, where on each trading day stocks are ranked by predicted trend scores and the top- $k$  are equally weighted, with realistic transaction costs applied (0.15% buy, 0.25% sell). Performance measured via Annual Return and Sharpe Ratio shows REST consistently outperforming alternatives in both returns and risk-adjusted metrics.

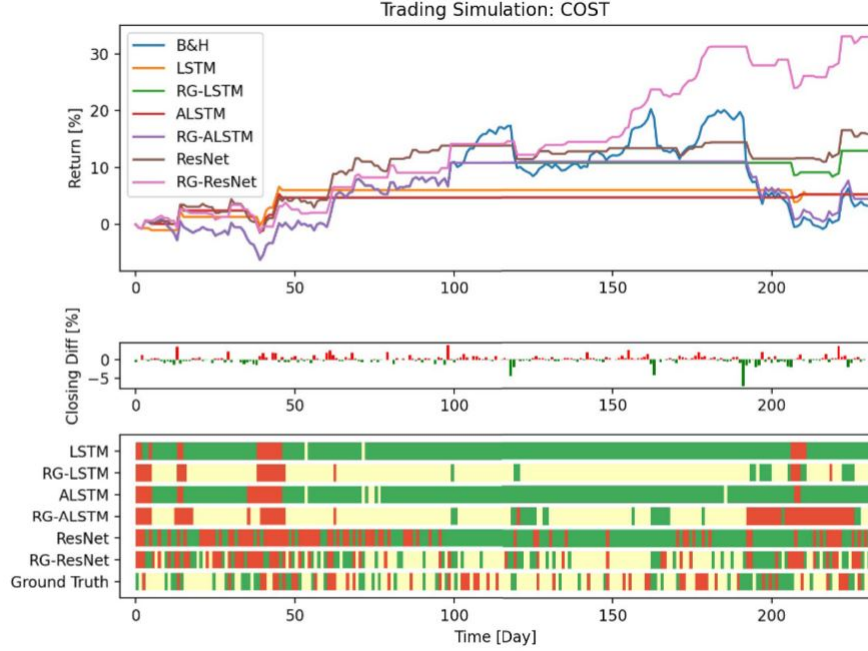


Figure 11: The cumulative return for Costco Wholesale Corp. (COST) from Li et al. [83], along with daily predictions and actual closing differences. The RGStockNet framework predictions are colour-coded: green for Rise, yellow for Flat, and red for Drop.

Cheng et al. [38] develop a complete investment system integrating their predictive model into a mobile mini-app and a web platform. The system extracts event-driven signals from financial news via a knowledge graph, provides buy/sell signals linked to events, and supports portfolio construction with explanatory context. Their contribution highlights a real-time, interactive tool that goes beyond theoretical evaluation toward practical investment use.

**From academic prototypes to production trading systems.** A significant gap persists between academic prototypes and production-grade trading systems. Academic studies emphasise predictive accuracy in controlled settings using curated data and offline backtesting ([97]), while industrial systems must meet strict latency constraints, process evolving data streams, and integrate with execution, risk, and compliance infrastructures ([98]).

This gap is amplified for multimodal models, as textual and relational data are often asynchronous and noisy, complicating real-time fusion ([75]). Assumptions common in academic work, such as stable distributions and full modality availability, rarely hold

in practice, where systems must handle missing signals and regime shifts.

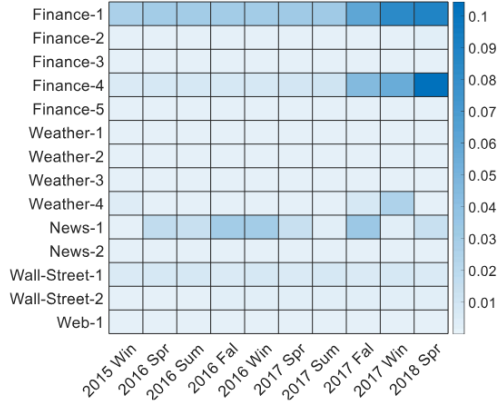
Beyond technical issues, regulatory and governance requirements, including explainability and traceability, further constrain deployment ([99]), often favouring modular and interpretable designs over highly entangled architectures.

## 6. Interpretability as a Cross-Cutting Property

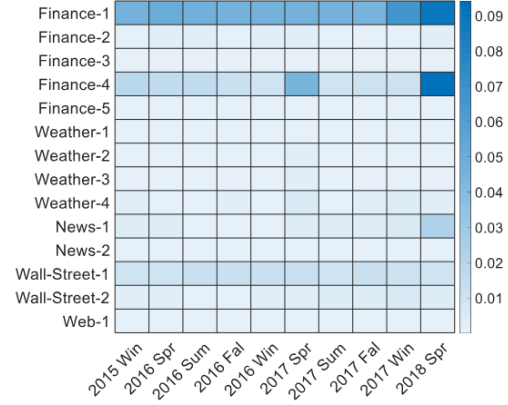
Building on the architectural and fusion choices introduced in Sections 3.2 and 3.3, and on the interpretability perspective formalised at the end of each fusion approach (Section 3.3), we analyse interpretability as an emergent property of multimodal model design rather than as a purely post-hoc add-on. In multimodal financial forecasting, interpretability is not tied to a specific model family or modality, but is shaped by how heterogeneous information is represented, integrated, and propagated through the model.

In financial applications, where prediction errors can lead to significant economic losses, interpretability is essential for transparency, accountability, and trustworthiness [100]. Recent work at the intersection of XAI and finance (e.g., Du et al. [101], Martins et al. [102], Jiang et al. [103]) emphasises understanding modality contributions, cross-modal interactions, and temporal dynamics. Interpretability supports risk management, model validation, and regulatory compliance [104], while helping distinguish meaningful patterns from spurious correlations in market data.

Fusion strategies play a central role in determining interpretability. Early fusion enables strong cross-modal interaction but often produces highly entangled representations, making it difficult to disentangle the contribution of market, textual, and relational signals. Intermediate fusion mechanisms partially alleviate this issue by preserving modality-specific encoders while enabling controlled interaction; attention- and gating-based strategies expose relevance weights across modalities, time steps, or graph neighbours, providing interpretable signals aligned with financial intuition, albeit without guarantees of causal faithfulness. Late fusion architectures offer the highest modality-level interpretability by maintaining independent processing pipelines and combining them only at the decision stage, thereby supporting traceability and auditability in regulated financial contexts.



(a) Attention heat map of Amgen.



(b) Attention heat map of Total System Services.

Figure 12: Attention heat map (the darker, the higher importance) from [77] between features and time steps according to the application of the Dandelion model to the two companies.

The Dandelion model [77] illustrates how interpretability can be embedded directly into multimodal architectures through its trinity attention mechanism across tasks, modalities, and time. As shown in Fig. 12, attention heat maps reveal how financial variables gain influence closer to the forecast horizon, while news-related features play a stronger role for Amgen, reflecting its higher public visibility. Similar principles are adopted in other attention-based models, such as THGNN [81], where interpretability is achieved by visualising attention weights over temporal and heterogeneous graph layers built on positive and negative price correlations. Overall, the literature indicates that interpretability in multimodal financial forecasting is best understood as a cross-cutting property emerging from architectural and fusion design choices rather than as an independent modelling dimension.

## 7. Open Challenges and Future Directions

Building on the insights of our taxonomy-driven review, we delineate persistent challenges in multimodal financial forecasting and propose a research roadmap that highlights standardization paths, evaluation practices, and open methodological gaps.

**Robustness to missing and misaligned modalities.** In operational environments, data streams are frequently incomplete, delayed, or out of sync. Developing architectures that degrade gracefully under such conditions remains a primary objective. Future

work should target principled handling of modality dropout and temporal misalignment, including uncertainty-aware training objectives and inference-time imputations that preserve calibration and reliability.

**Standardised and adaptive fusion.** Fusion practices range from naive concatenation to sophisticated attention mechanisms, yet the absence of shared evaluation protocols limits comparability. Research should pursue adaptive fusion schemes that weight modalities on the basis of data quality, recency, and task relevance, coupled with community benchmarks that enable fair, reproducible assessment across methods, markets, and horizons.

**Integrating LLMs within multimodal pipelines.** Large Language Models have the potential to significantly enrich textual channels by extracting higher-level semantic signals from news, social media, and financial disclosures, and by providing interpretable natural-language rationales for predictions. Recent multimodal financial models already rely on Transformer-based textual encoders tightly coupled with temporal and relational components, anticipating some of the design principles later enabled by general-purpose LLMs. However, the surveyed literature reveals that the explicit integration of LLMs as adaptive components within end-to-end multimodal pipelines remains limited. In most existing works, LLMs are used indirectly, for example as feature extractors or annotation tools, rather than as modules that dynamically interact with numerical time series or graph-structured data. This highlights an important research gap: designing multimodal architectures where LLMs actively participate in fusion and alignment mechanisms, enabling context-aware interaction across textual, temporal, and relational modalities while preserving robustness, interpretability, and quantitative consistency.

**Interpretability and decision support.** The composite nature of multimodal systems complicates post-hoc analysis. There is a pressing need for tools that attribute predictions to specific inputs and interactions across modalities and time. Such capabilities support risk management, model governance, and practitioner trust, and should be designed to operate under distribution shift and data sparsity.

**Domain adaptation and transfer.** Financial regimes vary across geographies, asset classes, and historical periods. Methods that adapt representations and decision rules to new domains, while controlling for drift and preserving performance guarantees, are essential for generalisability in heterogeneous and evolving settings.

**Broader and higher-quality modalities.** Augmenting models with well-curated alternative data—macroeconomic indicators, ESG signals, supply-chain events, and advanced sentiment measures—can expand the informational frontier. The emphasis should be on measurable incremental value, careful alignment and de-noising, and transparency about provenance and sampling biases.

**Benchmarks and datasets at scale.** Progress depends on accessible, large-scale benchmarks that reflect realistic constraints. Community resources should span multiple markets, asset classes, and modalities, include well-defined tasks and splits, and report costs as well as accuracy. Such artefacts will enable rigorous comparisons, ablations, and stress tests that assess robustness, scalability, and reproducibility. These challenges are further exacerbated by class imbalance and the prevalence of rare but high-impact events, which are intrinsic to financial markets yet are seldom addressed explicitly in multimodal benchmark design and evaluation.

Collectively, these directions chart a path toward multimodal systems that are more reliable under real-world constraints, more transparent in their reasoning, and more transferable across regimes and use cases.

## 8. Conclusion

This survey has presented a structured and comprehensive analysis of multimodal approaches for financial forecasting, covering developments from 2018 to 2025. By proposing a unified taxonomy along four dimensions—input modalities, modeling architectures, fusion strategies, and predictive tasks—we provided a systematic framework to compare 35 representative works. Our review highlights that integrating heterogeneous data sources such as market signals, textual news, and relational graphs can

substantially enhance predictive performance, but also introduces new challenges, including temporal misalignment, modality imbalance, missing or noisy data, and limited cross-market generalization.

We also discussed interpretability as a critical dimension for practical deployment, particularly in the financial domain where transparency, accountability, and regulatory compliance are essential. Attention-based mechanisms and graph-based explanations have emerged as promising tools to uncover the contribution of different modalities and their interactions over time.

Looking forward, research in multimodal financial forecasting is likely to benefit from advances in adaptive fusion techniques, learning with incomplete modalities, and the integration of large language models and temporal graph neural networks. Furthermore, developing standardized benchmarks, robust evaluation protocols, and explainability frameworks will be essential to foster reproducibility and accelerate progress in this rapidly evolving field. By bridging methodological innovation with domain-specific constraints, future multimodal systems have the potential to deliver not only higher predictive accuracy but also greater interpretability and trustworthiness, ultimately supporting more informed and responsible financial decision-making.

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